BPM2_ESOFT QUIZ 4_S2026_GD

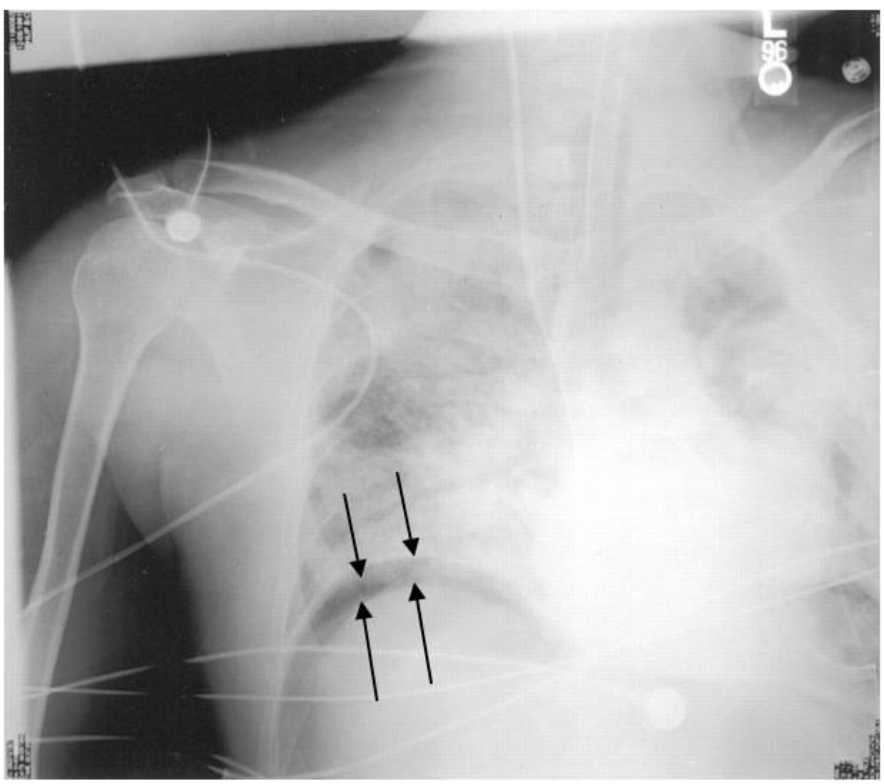
# of Questions: 30

Question #: 1

A 44-year-old man comes to the emergency department because of a 2-hour history of a puncture wound to the abdomen. Physical examination shows the anterior wall of the stomach is punctured (leading to the release of fluid and air into the abdominal cavity). A radiograph is taken and is shown in the attached image. In which of the following areas will you most likely find the collection of air in the abdomen of this patient?

A. Hepatoorenal recess
B. Paracolic gutters
✓C. Subphrenic recess
D. Lesser sac
E. Rectovesical pouch

Rationale: Whenever air is introduced into the abdominal cavity, this air will rise to the highest part of the cavity. This is the subphrenic recess.



Question #: 2

A 55-year old-man comes to the physician because of a 2-month history of severe abdominal pain that worsens after a fatty meal. He is scheduled for endoscopic retrograde cholangiopancreatography, where a fiberoptic scope is passed orally to the stomach, and into the small bowel. A cannula is then inserted into the major duodenal papilla in order to visualize the biliary tree. Which of the following locations will the tip of the endoscope most likely be found in order to enter the papilla?

A. 1st part of the duodenum
✓B. 2nd part of the duodenum
C. 3rd part of the duodenum
D. 4th part of the duodenum
E. Duodenojejunal junction

Rationale: The major duodenal papillae is located in the second part of the duodenum. This is an important landmark, as it is where the foregut ends and the midgut begins.

Question #: 3

A 45-year-old man is brought to the emergency department because of a 1-hour history of hematemesis. Physical examination shows severe pallor. His abdomen is distended and there are large dilated veins on the anterior abdominal wall. An anastomosis between which of the following venous structures is most likely responsible for the appearance of the patient's abdominal wall?

A. Left gastric and azygos
B. Superior and inferior rectal
C. Liver sinusoids and inferior phrenic
✓D. Superficial epigastric and paraumbilical
E. Left colic and renal

Rationale: This patient has cirrhosis of the liver which would lead to portal hypertension. One of the clinical signs of portal hypertension is caput medusa, which are dilated veins on the anterior abdominal wall. This is due to anastomosis between the para-umbilical veins (portal) with the superficial epigastric veins (caval).

Question #: 4

A 25-year-old woman is preparing for a 100-meter practice sprint. She is a sprinter who has a range of blood tests regularly performed to monitor her health, nutritional state, and adaptation to training. During one of the testing sessions her plasma glucose is 90 mg/dl (70-100 mg/dL), and her insulin/glucagon ratio is low. Which of the following best describes her metabolic state?

A. Phosphorylation and activation of glycogen synthase
✓B. Phosphorylation and activation of AMP-kinase by glucagon
C. Inhibition of adipocyte hormone-sensitive lipase
D. Inhibition of CPT-I due to high levels of malonyl CoA
E. Elevated hepatic fructose 2,6-bisphosphate levels
F. Low activity of mitochondrial HMG-CoA synthase

Rationale: The vignette is reviewing the metabolic changes when insulin: glucagon ratio is low (fasting);
Glycogen synthase is inactive and phosphorylated by glucagon - protein kinase A system; Glycogen phosphorylase is active in the phosphorylated state
Glucagon activates protein kinase A (increased cAMP levels), which phosphorylates AMP-kinase and activates it. AMP-kinase adds phosphate groups to acetyl CoA carboxylase (fatty acid synthesis) and HMG-CoA reductase (cholesterol synthesis) and inhibits both processes.
Adipocyte hormone-sensitive lipase is active by phosphorylation and fatty acids are released from adipose tissue and are used by liver and muscle for beta-oxidation.
CPT-I is active due to low cytosolic malonyl CoA levels. This results in increased transport of fatty acids into mitochondria for beta-oxidation.
Hepatic fructose 2,6-bisphosphate levels are low due to increased activity of fructose 2,6 bisphosphatase (FBPase-2) of the bifunctional enzyme; Low levels of fructose 2,6-bisphosphate activates fructose 1,6-bisphosphatase of gluconeogenesis; and inhibits PFK-1 of glycolysis.
Mitochondrial HMG-CoA synthase activity is increased in the fasting state and there is increased formation of ketone bodies

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Question #: 5

A 10-month-old girl is brought to the physician for a follow-up examination. She was diagnosed with MCAD deficiency by newborn screening which was followed by genetic testing. She is advised to be fed every 6-8 hours in order to prevent fasting hypoglycemia.Which of the following hepatic enzymes is most likely lacking an allosteric activator during fasting in this patient?

A. Glycogen phosphorylase
✓B. Pyruvate carboxylase
C. Phosphoenolpyruvate carboxykinase (PEPCK)
D. Alanine aminotransferase (ALT)
E. Acetyl CoA carboxylase
F. Pyruvate dehydrogenase (PDH) complex

Rationale: Patients with MCAD deficiency present with hypoglycemia and hypoketosis due a metabolic crisis.
MCAD deficiency results in impaired beta-oxidation and hence reduced formation of acetyl CoA from fatty acid oxidation.
Acetyl CoA is an absolute activator of pyruvate carboxylase (gluconeogenesis) and hence there is low activity of gluconeogenesis. Also, there is less ATP formed which is is not sufficient to fuel gluconeogenesis. As a result, gluconeogenesis is impaired and patients present with severe fasting hypoglycemia.
The precursor of ketone bodies is acetyl CoA; Impaired beta-oxidation in MCAD deficiency results in less acetyl CoA availability for formation of ketone bodies.

Question #: 6

A 2-year-old boy is brought to the physician by his mother because of a 3-month history of tripping over objects and developmental delay. Physical examination shows milestone developmental delays. Examination of the skin punch biopsy specimen shows abnormal lysosomes. Which of the following compounds is most likely accumulating in this patient's lysosomes?

A. Sphingomyelin
B. Globoside
✓C. Ganglioside
D. Glycosaminoglycan
E. Glycogen
F. N-linked glycoprotein
G. O-linked glycoprotein

Rationale: The vignette describes a patient with Tay-Sachs disease. The enzyme deficient is lysosomal hexosaminidase A. Characterized by developmental delay and blindness (macular cherry red spot).
Lysosomal accumulation of gangliosides results in the 'onion-shell' appearance. Gangliosides are glycolipids.

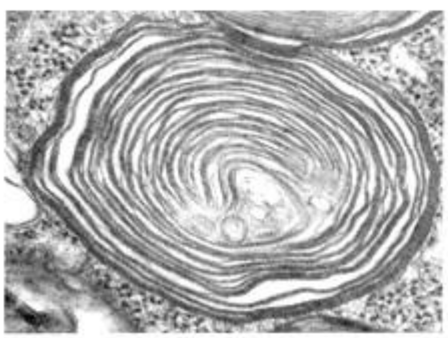
Sphingomyelin accumulation is a feature of Niemann-Pick disease. Enzyme deficient is sphingomyelinase. Characterized by blindness (cherry red spot in macula), developmental delay and hepatomegaly. Sphingomyelin is phospholipid.

Fabry disease is characterized by globoside accumulation. Deficient enzyme is alpha-galactosidase A.

Gaucher disease is characterized by deficiency of glucocerebrosidase and accumulation of glucocerebrosides. Characterized by bone pain and anemia.

Lysosomal accumulation of glycogen is characteristic of Pompe disease. Deficient enzyme is lysosomal acid maltase.

Hurler (more severe) and Hunter (less severe) syndromes are characterized by accumulation of glycosaminoglycans in lysosomes. Characterized by skeletal deformities, hepatosplenomegaly, developmental delay, corneal clouding (Hurler syndrome). Hurler is deficiency of iduronidase enzyme and Hunter is deficiency of iduronate sulfatase enzyme. Both are characterized by accumulation of dermatan and heparan sulfate (glycosaminoglycans).



Question #: 7

A 20-year-old woman is brought to the emergency department by her friends in an unconscious state. Her respirations are 30/min and are deep and rapid. Her breath test is positive for ethanol. Acid base studies show metabolic acidosis. The high NADH/ NAD+ ratio in hepatocytes is most likely responsible for her metabolic state. Which of the following reactions is most likely favored in her liver?

✓A. Oxaloacetate to malate
B. Pyruvate to acetyl CoA
C. Pyruvate to oxaloacetate

D. Oxaloacetate to phosphoenolpyruvate
E. Lactate to pyruvate

Rationale: Binge ethanol consumption results in excessive ethanol oxidation via alcohol dehydrogenase and aldehyde dehydrogenase. This increases the NADH/NAD+ ratio.
This favors the trapping of oxaloacetate as malate. Oxaloacetate is not available for gluconeogenesis.
Pyruvate is converted to lactate. This results in lactic acidosis. Pyruvate is not available for gluconeogenesis.
The precursors of gluconeogenesis (pyruvate and oxaloacetate) are shunted away from gluconeogenesis. This results in hypoglycemia.

Question #: 8

An 18-year-old man comes to the physician because of a 2-week history of skin lesions. He has been on a special diet containing raw egg whites for the past 3 months. Avidin present in raw eggs inhibits the absorption of biotin. Which of the following reactions is most likely impaired in this patient?

A. Conversion of homocysteine to cystathionine
B. Conversion of methylmalonyl CoA to succinyl CoA
C. Synthesis of thymidine mono phosphate
D. Conversion of phenylalanine to tyrosine
E. Synthesis of serotonin
✓F. Conversion of propionyl CoA to methylmalonyl CoA

Rationale: Biotin (vitamin B7) is required for carboxylation reactions: Pyruvate carboxylase (gluconeogenesis); Propionyl CoA carboxylase (conversion of propionyl CoA to methyl malonyl CoA) and acetyl CoA carboxylase (regulated enzyme of fatty acid synthesis; converts acetyl CoA to malonyl CoA).

Conversion of homocysteine to cystathionine requires PLP (Vitamin B6). Cystathionine synthase requires PLP.

Conversion of methylmalonyl CoA to succinyl CoA requires vitamin B12 (cobalamin).

Synthesis of thymidine mono phosphate (pyrimidine nucleotide) and purine nucleotides requires folic acid (vitamin B9).

Phenylalanine hydroxylase which converts Phenylalanine to Tyrosine requires tetrahydrobiopterin.

Serotonin synthesis requires tetrahydrobiopterin (tryptophan hydroxylase) and PLP (5-hydroxy tryptophan decarboxylase).

Question #: 9

A 34-year-old man comes to the physician because of a 3-day history of painful skin blisters. A week earlier, he had been on a Caribbean vacation and sunbathed and drank more alcohol than usual. He developed blisters on the skin and later avoided sun exposure. His urine is red on voiding. Urinalysis is performed. Which of the following molecules would most likely be found in the highest concentrations in his urine?

✓A. Uroporphyrin III
B. Porphobilinogen
C. Coproporphyrin I
D. Coproporphyrin III
E. Protoporphyrin IX
F. Bilirubin
G. Homogentisic acid

Rationale: The vignette describes a patient with porphyria cutanea tarda. Note the age of the patient and the presence of photosensitivity. There is deficiency of Uroporphyrinogen decarboxylase.
Congenital erythopoietic porphyria presents very early (infants or young children) and there is photosensitivity.
Acute intermittent porphyria usually presents in adolescents. Presents with abdominal pain and no photosensitivity.

Question #: 10

A 21-year-old woman comes to the physician because of a 2-month history of intermittent epigastric pain. Physical examination shows mild scleral icterus. There is no hepatosplenomegaly. Results of laboratory studies are shown in the table. Which of the following best explains the findings in this patient?

A. Decreased activity of UDP-glucuronyl transferase
B. Gall stones obstructing the common bile duct
✓C. Defect in the ABC transporter which secretes conjugated bilirubin

D. Increased rate of hemolysis
E. Hepatocellular damage due to viruses

Rationale: The lab findings show an increase in direct (conjugated) bilirubin and normal AST, ALT and ALP levels. These are indicative of Dubin-Johnson syndrome in which there is a defect in the ABC-transporter that secretes conjugated bilirubin into the biliary canaliculus.

	Patient	**Reference range**
Serum total bilirubin	3.5 mg/ dL	0.1-1.0 mg/dL
Serum direct bilirubin	3.2 mg/dL	0.1-0.3 mg/dL
Serum Aspartate aminotransferase (AST)	20 U/L	8-20 U/L
Serum Alanine aminotransferase (ALT)	20 U/L	8-20 U/L
Serum Alkaline phosphatase (ALP)	60 U/L	20-70 U/L
Urine bilirubin	Present	Absent

Question #: 11

A 10-year-old boy is brought to the physician by his parents because of a 2-day history of swelling of his feet. His mother says he may have accidentally drunk automobile antifreeze four days ago. Physical examination shows bilateral pedal edema and puffiness of his face. Laboratory studies show increased serum creatinine and increased blood urea nitrogen levels. Arterial pH is low and there is increased anion gap metabolic acidosis. Which of the following metabolites is most likely responsible for the findings in this patient?

A. Ethanol
B. Methanol
C. NAPQI
D. Acetaminophen
E. Salicylates
✓F. Oxalate
G. Formate
H. Acetaldehyde

Rationale: The boy has ethylene glycol poisoning and renal failure due to excessive formation of oxalate. Oxalate is an unmeasured anion and increases the anion gap. Serum Bicarbonate levels will be low (metabolic acidosis)
Formate and formaldehyde are formed from methanol. These intermediates affect vision and may result in blindness. Patients with methanol poisoning may also have high anion gap metabolic acidosis. Formate is an unmeasured anion and serum bicarbonate levels are low

lowering arterial pH (High anion gap metabolic acidosis)

Question #: 12

A 35-year-old woman is brought to the emergency department by her husband in an unconscious state. Results of laboratory studies on admission are shown in the table. Which of the following is most likely responsible for this clinical presentation?

A. Diabetic ketoacidosis
B. Hypoglycemia due to insulin overdose
C. Hyperosmolar hyperglycemic state
✓D. Prolonged starvation
E. Adrenergic-corticoid phase following an accident

Rationale: An important complication of prolonged starvation is hypoglycemia. A patient with diabetic ketoacidosis has elevated blood glucose levels, low arterial pH, elevated ketone bodies and very low (undetectable) insulin levels.

Parameter	Patient	Reference range
Blood glucose	55	60-100 mg/dL
Ketone bodies	15	<1 mg/dL
Arterial pH	7.35	7.38-7.42
Insulin levels	5	5-25 µU/ml

Question #: 13

Laboratory and metabolic studies are performed in a 30-year-old man who has been starving for 10 days and a 30-year-old man with critical illness. Which of the following findings will most likely be observed in these individuals?

A. I
B. II
C. III
D. IV
✓E. V

Rationale: In the flow phase of critical illness, there is elevated cortisol, higher metabolic rate and high levels of insulin secretion (insulin resistance) and elevated blood glucose levels. The ketone body levels are usually low following critical illness; whereas they are elevated in prolonged starvation.

		Starvation	Critical illness
I	Cortisol levels	High	Low
II	Metabolic rate	High	Low
III	Insulin levels	Low	Low
IV	Ketone body levels	Low	Very high
V	Blood glucose levels	Normal	High

Question #: 14

A 25-year-old man who is a downhill slalom skier is brought to the emergency department because of an accident during skiing. Corrective orthopedic surgery was performed for compound fractures to his right femur and right tibia on the day of the accident. He also has several broken ribs. There is no evidence of internal injuries. Following surgery, the patient is transferred to a post-operative ward for observation. Which of the following statements would be most accurate regarding the man's status five days after the initial accident?

A. Decreased C-reactive protein levels
✓B. Negative nitrogen balance
C. Decreased urinary urea
D. Decreased PGE_2 levels

E. Decreased adipocyte lipolysis

Rationale: The man is in the 'Flow' phase following critical illness which is characterized by muscle wasting (negative nitrogen balance) and increased lipolysis. There would be higher excretion of urea in urine. The levels of inflammatory proteins (C-reactive proteins) and inflammatory mediators (PGE2) would be elevated.

Question #: 15

A newborn is seen by the neonatologist for recurrent severe diarrhea every time he breastfeeds. A significant amount of H_2 is measured in his breath after breastfeeding. Which of the following enzyme deficiencies is the most likely diagnosis in this patient?

✓A. Lactase deficiency
B. Fructokinase deficiency
C. Aldolase B deficiency
D. Alpha-Amylase deficiency
E. Galactose 1-phosphate uridyl transferase deficiency
F. Galactokinase deficiency
G. Glucose 6-phosphatase deficiency

Rationale: The vignette describes a child with congenital lactose intolerance due to deficiency of intestinal lactase. This is differentiated from galactosemia in which the children have liver manifestations, cataracts and developmental milestone delay.

Question #: 16

A 33-year-old man is brought to the physician by his wife because of a 1-year history of anorexia and weight loss. His wife says that he has mild behavioral changes over the same period. Physical examination shows that the liver is not palpable. Eye examination shows Kayser-Fleischer (KF) rings in the cornea. Examination of a liver biopsy specimen shows cirrhosis. Which of the following best accounts for the clinical presentation?

✓A. Defect in biliary copper excretion
B. Defect in dietary copper absorption
C. Increase in dietary iron absorption
D. Deficiency of dietary iron uptake

E. Decreased intrinsic factor secretion from parietal cells
F. Increased incorporation of copper to ceruloplasmin
G. Homozygosity for mutation in the *HFE* gene

Rationale: The vignette describes a patient with Wilson disease: Liver damage, behavioral changes and corneal ring (K-F ring). Wilson disease is due to a defect in excretion of copper from the liver due to a mutation in the gene encoding a copper transport ATPase. Also note that there is impaired incorporation of copper into ceruloplasmin resulting in low serum ceruloplasmin levels.

Question #: 17

A 38-year-old man comes to the physician because he is finding it difficult to lose weight. He feels that he is gaining more weight despite dietary modifications. His BMI is 38 kg/m^2, and waist-to-hip ratio is 1.3. Which of the following hormonal conditions is most likely contributing to this patient's uncontrolled weight gain?

A. Increased adiponectin secretion
B. Adiponectin resistance
C. Resistance to growth hormone
✓D. Leptin resistance
E. Elevated cortisol secretion

Rationale: Leptin is an adipokine secreted from adipocytes. Leptin levels are increased when a person gains weight. Leptin reduces appetite and increased energy expenditure. However, in an obese individual the high levels of leptin may lead to leptin resistance where the target tissues are not responsive to the increased leptin levels. Leptin resistance is one of the contributory factors to the development of insulin resistance. Adiponectin levels decrease following weight gain.

Question #: 18

A 35-year-old man comes to the physician because of a 3-day history of malaise and yellowing of the skin. Physical examination shows scleral icterus. Results of laboratory studies are shown in the table. Which of the following is the most likely diagnosis?

A. Obstruction of the common bile duct due to pancreatic cancer
B. Gilbert syndrome

C. Chronic alcoholic cirrhosis
D. Hemolysis due to sickling crisis
✓E. Acute hepatocellular damage due to viruses
F. Dubin-Johnson syndrome

Rationale: The laboratory tests are suggestive of acute hepatitis: The Elevated levels of ALT and AST; Higher ALT levels than AST levels; An increase in both indirect and direct bilirubin (mixed hyperbilirubinemia); Normal albumin levels indicate acute hepatitis. Biliary obstruction would most likely result in highly elevated ALP and GGT levels, and higher levels of direct bilirubin than indirect bilirubin.

Serum parameter	Patient	Reference range
Aspartate aminotransferase (AST)	800	8-20 U/L
Alanine aminotransferase (ALT)	1050	8-20 U/L
Alkaline phosphatase (ALP)	150	20-70 U/L
γ-glutamyl transferase (GGT)	70	10-30 U/L
Direct bilirubin	3.8	0-0.3 mg/dL
Total bilirubin	7.4	0.1-1.0 mg/dL
Albumin	4.2	3.5-5.5 g/dL

Question #: 19

A 2-year-old girl is brought to the physician by her parents because of a 2-month history of failure to thrive and diarrhea. Physical examination shows skin lesions in the exposed areas of the neck, hands, and feet. Urinalysis shows the presence of tryptophan. Which of the following is the best explanation for the physical examination findings in this patient?

A. Negative nitrogen balance
B. Tryptophan inhibiting serotonin synthesis
C. Tryptophan inhibiting melanin synthesis
✓D. Decreased NAD^+ synthesis from tryptophan
E. Accumulation of homocysteine and methylmalonate

Rationale: The vignette describes a patient with Hartnup disease due to a defect in the tryptophan transporter in GIT and in the renal tubules. Tryptophan is a precursor of NAD. Tryptophan deficiency in these patients results in reduced NAD formation from tryptophan and may result in manifestations of pellagra (dermatitis, diarrhea and neurological features).

Question #: 20

A 4-year-old girl is brought to the physician by her father because of a 2-month history of vision problems. She is taller than age and sex-matched controls. Physical examination shows lens dislocation. There are also early signs of atherosclerosis. Administration of pyridoxine results in improvement. Which of the following processes is most likely affected in this patient?

A. Catabolism of the branched chain amino acids
B. Beta oxidation of fatty acids
✓C. Conversion of homocysteine to cysteine
D. Synthesis of heme from glycine and succinyl CoA
E. Liver glycogenolysis

Rationale: The vignette describes a child with homocystinuria (deficiency of cystathionine synthase). It is characterized by lens dislocation and premature atherosclerosis.

Question #: 21

A 4-month-old boy is brought to the physician by his parents because of failure to thrive since birth. He is short in stature compared to age and sex-matched controls. Physical examination shows coarse facial features, macrocephaly and hepatosplenomegaly. His joint mobility is severely restricted and there is corneal clouding. Examination of skin biopsy specimens show the presence of cytosolic inclusion bodies in the fibroblasts. Which of the following is most likely responsible for the clinical presentation in this patient?

A. Isolated defect in the degradation of gangliosides
B. Isolated deficiency of sphingomyelinase
C. Defect in the synthesis of glycosaminoglycans
✓D. Defect in sorting of lysosomal enzymes
E. Defect in the synthesis of sphingoglycolipids

Rationale: The vignette describes a child with I-cell disease. The defect in I-cell disease is that there is a defect in the sorting of lysosomal enzymes due to a defect in phosphorylation of mannose residues. As a result, the lysosomal enzymes are secreted from the cell rather than being targeted to the lysosomes. This is the most severe of the lysosomal storage disorders and there is a defect in all lysosomal enzymes and hence an accumulation of complex lipids (glycolipids, sphingolipids) and glycosaminoglycans.

Question #: 22

Fifty women aged group of 35 to 40 years are recruited as volunteers for the study of digestion and absorption. They are given a meal containing 40% fat. Which of these molecules is best suited for enhancing digestion and subsequent absorption of this macronutrient by the intestinal mucosal cells?

A. Deoxycholic acid
B. Chenodeoxycholic acid
C. Cholic acid
✓D. Taurocholic acid
E. Lithocholic acid
F. Bilirubin diglucuronide

Rationale: The conjugated bile acids/salts assist in lipid digestion and absorption. Chenodeoxycholic acid and cholic acid are conjugated with glycine or taurine to form the conjugated bile acids/salts which are glyco or taurocholic acid and glyco or taurochenodeoxycholic acid. Deoxycholic acid and lithocholic acid are secondary bile acids which are lost in feces. Bilirubin diglucuronide (conjugated bilirubin) is also known as bile pigment and is formed by the breakdown of heme.

Question #: 23

A 13-year-old girl is brought to the physician by her parents because of a 2-day history of increased frequency of urination, increased thirst and weight loss. Four days ago, she was diagnosed with a lung infection and was started on antibiotics. She responded poorly to the antibiotics and the infection did not improve. Physical examination shows dry mucus membranes. Urine dipstick tests are positive for glucose and ketone bodies. Which of the following is most likely an additional laboratory finding in this patient?

✓A. Very low circulating insulin levels
B. Decreased plasma glucose levels
C. Normal HbA1c levels
D. Decreased glucose output by liver
E. Increased circulating C-peptide levels

F. Increased arterial pH
G. Increased serum bicarbonate levels

Rationale: The vignette describes a patient with type I diabetic ketoacidosis which is characterized by very low or undetectable circulating insulin levels and C-peptide levels.

Question #: 24

A 39-year-old man is brought to the emergency department because of a 4-hour history of severe abdominal pain. He has consumed excessive amounts of alcohol over the past 2 days. Physical examination shows acute tenderness to his epigastric region. Laboratory studies show elevated levels of pancreatic enzymes suggestive of acute pancreatitis. This condition is characterized by autolysis of the secretory units of the parenchyma of this gland as a result of the activity of a stimulatory hormone. Which of the following cells most likely secretes the stimulatory hormone?

A. Parietal
B. Paneth
C. Chief
✓D. Enteroendocrine
E. Goblet
F. Mucous

Rationale: Cholecystokinin (CCK is a polypeptide hormone secreted by the enteroendocrine cells of the duodenum that causes the acinar cells of the exocrine pancreas to secrete their proenzymes.

Question #: 25

A 25-year-old man comes to the physician because of a 1-month history of intermittent abdominal cramps and bloating. Physical exam shows no abnormalities. History shows the symptoms typically appear following a meal rich in carbohydrates. Genetic testing shows a mutation in a gene for an enzyme that breaks down oligosaccharides to monosaccharides for absorption. Which of the following best characterizes the localization of this enzyme?

A. Digestive organelles of hepatocytes
B. Cytoplasm of enteroendocrine cells

✓C. Microvilli of enterocytes
D. Glands of the intestinal submucosa
E. Lumen of the stomach

Rationale: Enterocytes are the major cell type found in the epithelial lining of the intestinal tract. They produce enzymes that are inserted into the apical plasma membrane that function in the terminal digestion and absorption, which remains their primary function.

Other cells of the intestinal epithelium include paneth cells, which produce lysozymes etc, goblet cells which produce mucus, enteroendocrine cells which produce ghrelin, secretin etc..

Question #: 26

A 43-year-old man comes to the physician because of a 2-week history of diarrhea and abdominal pain. He has a 3-year history of heavy alcohol consumption. Physical examination shows tenderness of the epigastric region. Laboratory studies of the blood lead to the diagnosis of chronic pancreatitis, a condition characterized by the slow autolytic destruction of the parenchyma by its own enzymes. Which hormone regulates bicarbonate secretion from this gland?

A. Gastrin
✓B. Secretin
C. Cholecystokinin
D. Motilin
E. Glucagon
F. Ghrelin

Rationale: Enterochromaffin (EC) cells are examples of minor cell types in pancreatic islets (also located in pancreatic exocrine acini and duct epithelium) that secrete Secretin (option b), Motilin and Substance P. Secretin acts locally on the pancreas to stimulate bicarbonate secretion in pancreatic juice.
a) Gastrin- stimulates secretion of gastric juices
c) Cholecystokinin- causes pancreatic acinar cells to secrete their proenzymes
d) Motilin- increases gastric and intestinal motility
e) Glucagon- increases blood glucose levels via gluconeogenesis and glycogenolysis
f) Ghrelin- stimulates appetite

Question #: 27

A 58-year-old man comes to the physician because of a 2-month history of heart burn and belching. Laboratory studies show that his gastric acid output exceeds normal levels. Ultrasonography of the abdomen shows a 2.3-cm mass in the pancreas. He is suspected of having a tumor in the pancreas. This tumor secretes a large amount of gut hormone that is responsible for increased gastric acid secretion in the stomach. Which of the following gastro-intestinal cell types is most likely the source of this hormone?

A. S cells
✓B. G cells
C. Chief cells
D. D cells
E. Enteroendocrine cells

Rationale: This patient has a pancreatic tumor that secretes large amounts of the hormone, gastrin. The source of gastrin is G cells in the GI-tract.
S cells: release secretin (pancreatic HCO_3^-)
G cells: release gastrin -> gastric acid upregulation - correct answer
Chief cells: release digestive enzymes (pepsin & lipase)
D cells: release somatostatin, inhibits gastric acid release
Enteroendocrine cells: generic grouping of all cells that release GI hormones, much less specific answer than G cells

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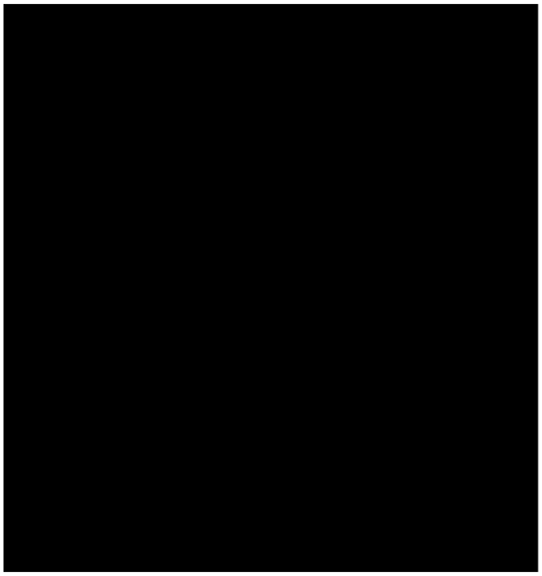
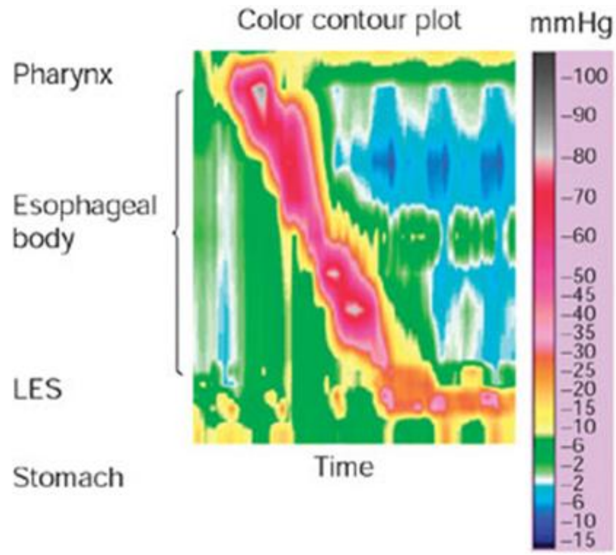
Question #: 28

A 48-year-old man comes to the physician because of a 3-week history of diffuse chest pain and difficulty swallowing. His vital signs are within normal limits. Physical examination shows no abnormalities. The result of high-resolution esophageal manometry is shown in the attached color contour plot. Which of the following interpretations of this manometry result is most accurate?

A. The upper esophageal sphincter does not open
B. Primary peristalsis is absent in the esophageal body
C. The lower esophageal sphincter does not open
✓D. There is functional peristalsis in the upper and lower esophagus
E. Secondary peristalsis is absent in the esophageal body

Rationale: Signs of transient high pressure (red colour) are seen in the UES and LES, which implies that both sphincters are opening and closing. The transient pressure increase travelling down the esophageal body in the correct temporal sequence also shows that peristalsis is present and functional in the upper and lower esophagus. The achalasia in this patient is unlikely to be related to esophageal motor dysfunction.





Question #: 29

A 32-year-old man comes to the physician because of a 6-month history of recurrent episodes of burning sensation in the throat. He says that these episodes usually wake him up at night. His BMI is 34 kg/m^2 (Normal: 19-25 kg/m^2). Physical examination shows no abnormalities. Endoscopy of the upper GIT shows peptic ulcers in the proximal duodenum. He is advised to take a drug that inhibits H^+/K^+ ATPase in the stomach for 4 weeks. Which of the following effects is most likely expected in this patient after a meal?

✓A. Failure to increase pH of the gastric venous blood
B. Increased carbonic anhydrase activity in the stomach
C. Decreased Na^+ in gastric juice
D. Increased H^+ in gastric juice
E. Gastric juice becomes hypotonic

Rationale: The H+/K+ ATPase in gastric parietal cells is the key mechanism for stomach acidification after a meal, which usually results in a temporary increase in gastric venous blood pH, because parietal cells secrete HCO3- secrete into the bloodstream to compensate for the H+ release into the stomach. Because the drug inhibits the H+/K+ ATPase, it would also reduce this post-meal alkalinity of gastric venous blood. Hence, there is failure to increase pH of the gastric venous blood.

Question #: 30

A 33-year-old man is brought to the emergency department by his friend because of a 1-hour history of altered consciousness. His friend says he was participating in a hunger strike for the past 3 days and did not eat or drink anything. His pulse is 104/min and blood pressure is 104/62 mm Hg. Physical examination shows dry mucous membranes, sunken eyes and altered mental state. Abdominal auscultation shows occasional bowel sounds. These sounds are produced by strong propulsive contractions that pass from stomach through small intestine during fasting. Which of the following humoral substances most likely mediates this intestinal motility in the patient?

A. Acetylcholine
✓B. Motilin
C. Nitric Oxide
D. Vasoactive Intestinal polypeptide
E. Serotonin

Rationale: Fasting generates migrating motor complex (MMC), which sweeps and empties the intestines during longer fasting periods. The key inducer of the MMC in the fasted state is motilin, hence this is the correct answer.